



# Global Currency Market Intelligence Dashboard

## Problem Statement

International businesses, import/export companies, traders, and finance teams rely on accurate exchange-rate information to make informed financial decisions. Manually collecting exchange rates from multiple websites is time-consuming, inconsistent, and difficult to track historically.

This project addresses that challenge by building a fully automated data pipeline that collects, stores, and visualizes historical exchange-rate data in an interactive Power BI dashboard, enabling users to monitor currency movements and analyze trends efficiently.

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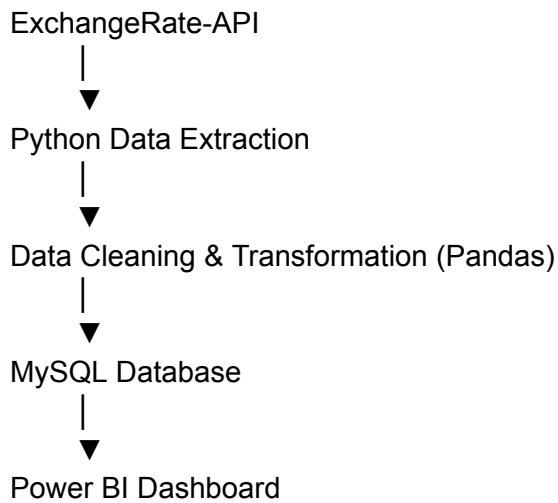
## Project Objectives

- Automate daily exchange-rate data collection without manual intervention.
  - Build and maintain a historical exchange-rate database.
  - Develop an interactive Power BI dashboard for country-wise currency analysis.
  - Monitor daily exchange-rate movements and historical trends.
  - Create a scalable analytics solution that can be extended to forecasting and advanced financial analysis.
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## Technology Stack

- **Python** – API integration and automation
  - **ExchangeRate-API** (<https://www.exchangerate-api.com/>) – Live exchange-rate data source
  - **Pandas** – Data cleaning and transformation
  - **MySQL** – Historical data storage
  - **Power BI** – Interactive dashboard development
  - **DAX** – Business calculations and measures
  - **Windows Task Scheduler** – Automated daily data refresh
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# ETL Workflow



## How the System Works

1. Python connects to the **ExchangeRate-API** and retrieves the latest exchange rates for **29 currencies** using **USD** as the base currency.
  2. The extracted JSON data is transformed into a structured tabular format using **Pandas**.
  3. Cleaned data is appended to a **MySQL** database, creating a continuously growing historical dataset.
  4. **Windows Task Scheduler** executes the Python script automatically every day, ensuring the database stays up to date.
  5. **Power BI** connects directly to MySQL and refreshes the dashboard using the latest available data and It Generate business insights such as current exchange rates, daily movement, country profiles, and historical trends.
  6. The result is a fully automated analytics solution requiring minimal manual effort.
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## Dashboard KPIs

- Latest Exchange Date
- Base Currency
- Total Countries Tracked
- Current Exchange Rate
- Daily Trend
- Selected Country
- Currency Code
- Exchange Summary (*Example: 1 USD = 94.97 INR*)

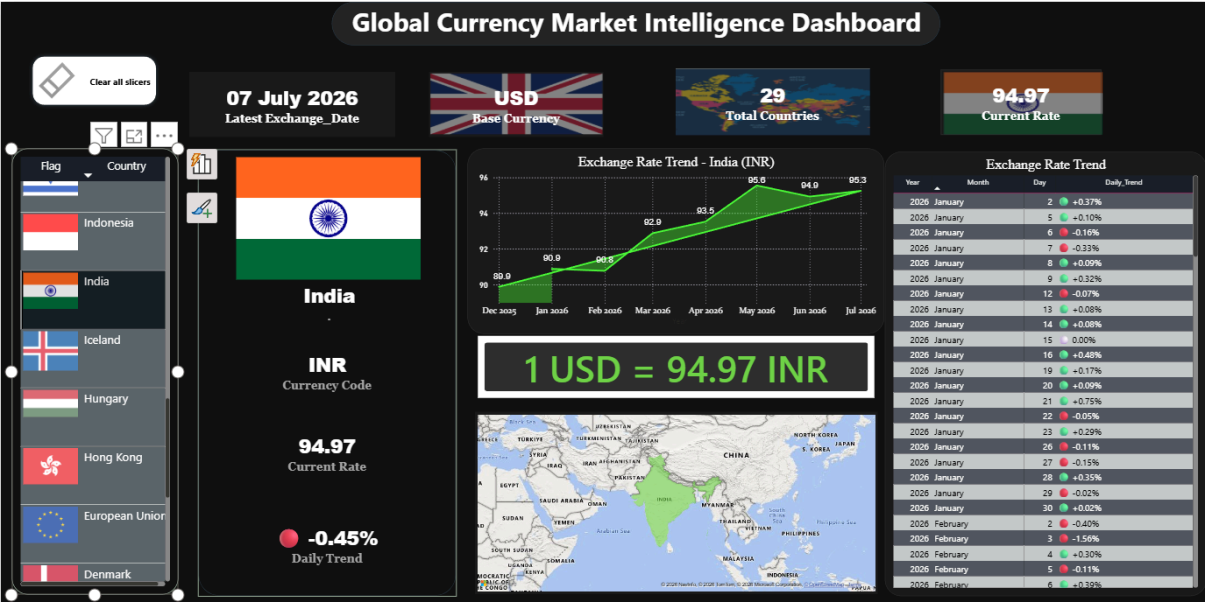
# Dashboard Features

- Country selector with national flags
- Dynamic country profile
- Exchange-rate trend visualization
- Historical exchange-rate analysis
- Interactive country map
- Dynamic chart titles
- Daily movement indicators
- Exchange summary card

# Business Impact

- Eliminated manual exchange-rate monitoring through automation.
- Built a historical exchange-rate repository for trend analysis.
- Enabled instant country-wise currency comparison.
- Improved accessibility to financial insights through an interactive dashboard.
- Established a scalable foundation for future predictive analytics.

# Dashboard Preview



# Conclusion

This project demonstrates a complete end-to-end data analytics workflow, beginning with automated API data extraction and ending with an interactive business intelligence dashboard. By integrating Python, MySQL, and Power BI, the solution transforms raw exchange-rate data into meaningful business insights through automation, historical analysis, and interactive visualization.